sample_ml

12 slides • made with Learnova

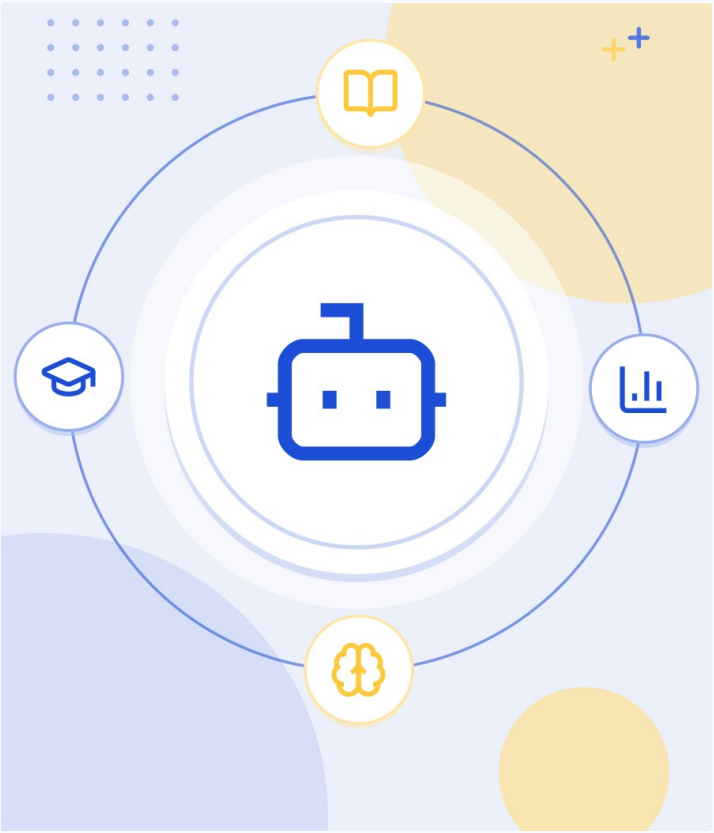


Machine Learning Basics: overview

Machine learning is a branch of artificial intelligence that lets computers learn patterns from data instead

Supervised learning uses labelled examples, such as emails marked spam or not spam, so the model

Overfitting happens when a model memorises the training data and fails on new data



Key terms: Machine learning — branch of artificial intelligence that lets computers learn patterns from data instead of following fixed rules · Training — adjusting those parameters to reduce an error measure called the loss · A model — mathematical function whose parameters are tuned using training data

Machine Learning Basics

Machine learning

A branch of artificial intelligence that lets computers learn patterns from data instead of following fixed rules

- A model is a mathematical function whose parameters are tuned using training data
- Training means adjusting those parameters to reduce an error measure called the loss

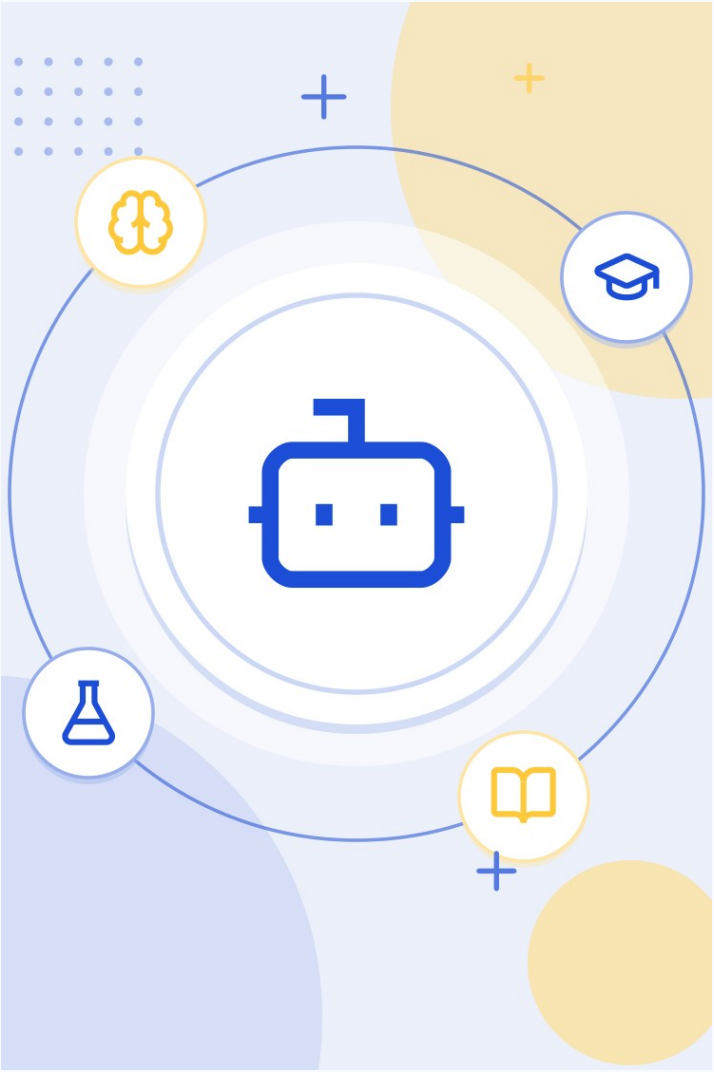
Key takeaway: **A branch of artificial intelligence that lets computers learn patterns from data instead of following fixed rules.**

Machine Learning Basics: Supervised learning uses labelled examples

Supervised learning uses labelled examples, such as emails marked spam or not spam, so the model can predict labels for new emails

Unsupervised learning has no labels and instead looks for structure, such as groups of similar customers

Reinforcement learning trains an agent by rewarding good actions and penalising bad ones



Checkpoint 1

QUESTION · RECALL

Fill in the blank: “Supervised learning uses labelled examples, such as emails marked spam or not spam, so the model can _____ labels for new emails.”

A repeats

B moves

C predict

D trains

Answer: C

“Supervised learning uses labelled examples, such as emails marked spam or not spam, so the model can predict labels for new emails.”

Machine Learning Basics: Overfitting happens when a model

Overfitting happens when a model memorises the training data and fails on new data

Detect it, we

To detect it, we hold back a validation set and compare the error on both sets

Regularisation adds a penalty for complexity and is one way to reduce overfitting

Cross validation repeats the train and validate split several times to give a more reliable estimate of performance



Key terms: **Training — adjusting those parameters to reduce an error measure called the loss · A model — mathematical function whose parameters are tuned using training data**

Gradient Descent

Gradient descent is an optimisation method that repeatedly moves the model parameters in the direction that reduces the loss

Learning rate controls

The learning rate controls the size of each step

Very large learning

A very large learning rate overshoots the minimum, while a very small one makes training slow

Key terms: **Training — adjusting those parameters to reduce an error measure called the loss**

Bias and Variance

Bias	Variance
Is the error from overly simple assumptions	Is the error from sensitivity to the particular training set
High causes underfitting	High causes overfitting
Good models balance the two, which is known as the -	Trade-off

Key takeaway: Key difference — Bias: Is the error from overly simple assumptions | Variance: Is the error from sensitivity to the particular training set
Key terms: Training — adjusting those parameters to reduce an error measure called the loss

Checkpoint 2

QUESTION · APPLY

Which statement describes Bias?

A. Machine learning: Machine learning is a branch of artificial intelligence that lets computers learn...

B. High causes underfitting

C. High causes overfitting

D. Is the error from overly simple assumptions

Answer: D

Bias: Is the error from overly simple assumptions. Variance: Is the error from sensitivity to the particular training set.

Evaluation Metrics

Accuracy is the share of predictions that are correct

Precision is the share of predicted positives that are truly positive

Recall is the share of actual positives that the model finds

F1 score

The F1 score is the harmonic mean of precision and recall



Glossary (1/2)

Machine learning

branch of artificial intelligence that lets computers learn patterns from data instead of following fixed rules

A model

mathematical function whose parameters are tuned using training data

Training

adjusting those parameters to reduce an error measure called the loss

Gradient descent

optimisation method that repeatedly moves the model parameters in the direction that reduces the loss

Bias

error from overly simple assumptions, while variance is the error from sensitivity to the particular training set

Accuracy

share of predictions that are correct

Glossary (2/2)

Precision

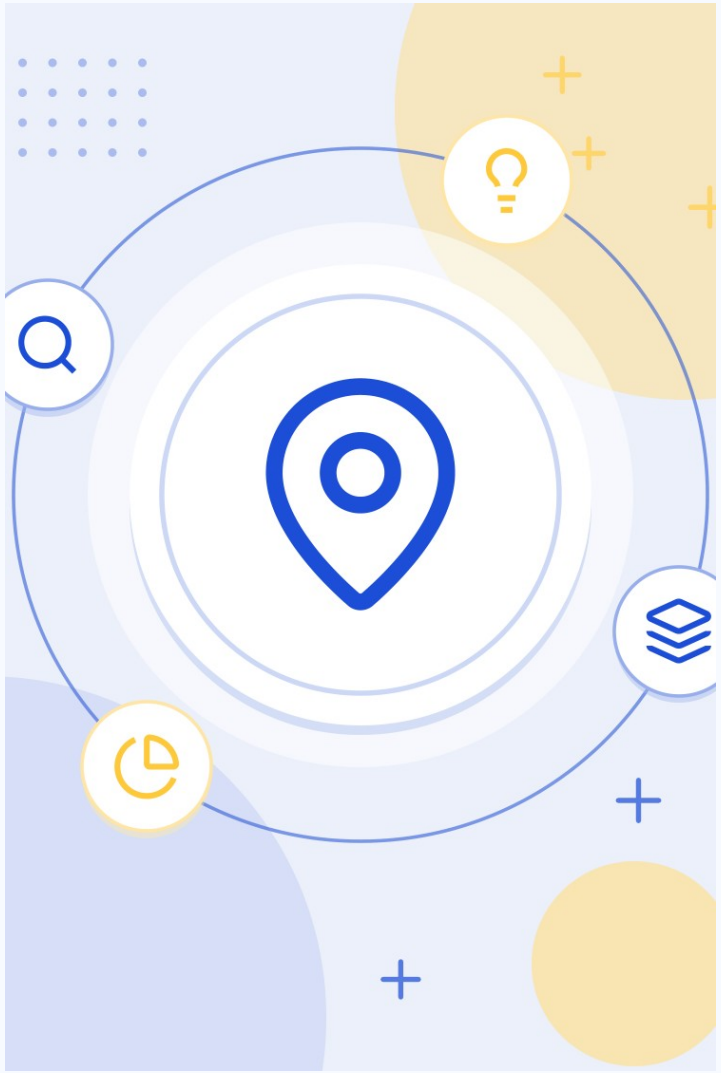
share of predicted positives that are truly positive

Recall

share of actual positives that the model finds

F1 score

harmonic mean of precision and recall



Checkpoint 3

QUESTION · RECALL

Fill in the blank: “_____: share of predicted positives that are truly positive”

A Precision

B overview

C assumptions

D emails

Answer: A

“Precision: share of predicted positives that are truly positive.”

Thank you

Questions?